Adaptive Heuristic Portfolio TSP Report

Overview

- Condition count: 8.
- Best held-out TSPLIB gap: phase8_best_single_fixed_heuristic.
- Best combined transfer gap: phase8_oracle_selector.
- Lowest selector regret: phase8_best_single_fixed_heuristic.
- Pareto-efficient conditions: phase8_full_solver_evolution, phase8_llm_static_selector, phase8_oracle_selector.

Run Metadata

- run_name: run_20260520_214625_m
- started_at_local: 2026-05-20 21:46:25
- finished_at_local: 2026-05-20 22:03:18
- duration_hhmm: 00:17
- duration_seconds: 1013.242
- seed_offset: 12000
- replicate_label: m
- judge_status: enabled
- judge_provider: openai
- judge_model: gpt-4.1-mini

Condition Comparison

Condition	Mode	Replay	Final TSPLIB Gap	Selector Regret	Runtime (ms)	Runtime-Adjusted Gap	Family Gap	Transfer Gap	Mean Novelty	Mean Complexity	Pareto Efficient
phase8_best_single_fixed_heuristic	best_fixed	none	0.07226	0.0	719.199271	0.07226	0.007284	0.031222	0.0	0.0	False
phase8_random_portfolio	random_portfolio	none	0.201486	0.129226	1712.379657	0.479729	0.126216	0.153947	0.0	0.0	False
phase8_oracle_selector	oracle_selector	none	0.07226	0.0	691.643686	0.07226	8.9e-05	0.026678	0.0	0.0	True

Condition	Mode	Replay	Final TSPLIB Gap	Selector Regret	Runtime (ms)	Runtime-Adjusted Gap	Family Gap	Transfer Gap	Mean Novelty	Mean Complexity	Pareto Efficient
phase8_supervised_ml_selector	supervised_selector	none	0.07226	0.0	704.474929	0.07226	0.007284	0.031222	0.0	0.0	False
phase8_llm_static_selector	static_selector	none	0.217506	0.145246	170.428671	0.217506	0.024668	0.095713	0.0	0.62	True
phase8_llm_evolved_adaptive_controller	adaptive_controller	none	0.086887	0.014627	1299.970086	0.15705	0.106009	0.098964	0.0	0.7175	False
phase8_llm_evolved_controller_diversity_failure_replay	adaptive_controller	diversity_failure	0.084733	0.012473	2029.476943	0.239104	0.090132	0.088143	0.647902	0.62	False
phase8_full_solver_evolution	full_solver	none	0.099035	0.026775	192.633443	0.099035	0.000881	0.037043	0.75	0.76	True

Condition Notes

phase8_best_single_fixed_heuristic

- Execution mode best_fixed with replay none.
- Final held-out gap 0.07226, selector regret 0.0, runtime-adjusted gap 0.07226.
- Family transfer 0.007284 and combined transfer 0.031222.
- Runtime 719.199271 ms, runtime inflation 0.0, mean novelty 0.0, and complexity 0.0.
- Pareto efficient: False.
- Fixed heuristic choice: farthest_insertion_two_opt.

phase8_random_portfolio

- Execution mode random_portfolio with replay none.
- Final held-out gap 0.201486, selector regret 0.129226, runtime-adjusted gap 0.479729.
- Family transfer 0.126216 and combined transfer 0.153947.
- Runtime 1712.379657 ms, runtime inflation 1.380953, mean novelty 0.0, and complexity 0.0.
- Pareto efficient: False.

phase8_oracle_selector

- Execution mode oracle_selector with replay none.
- Final held-out gap 0.07226, selector regret 0.0, runtime-adjusted gap 0.07226.

- Family transfer 8.9e-05 and combined transfer 0.026678.
- Runtime 691.643686 ms, runtime inflation -0.038314, mean novelty 0.0, and complexity 0.0.
- Pareto efficient: True.

phase8_supervised_ml_selector

- Execution mode supervised_selector with replay none.
- Final held-out gap 0.07226, selector regret 0.0, runtime-adjusted gap 0.07226.
- Family transfer 0.007284 and combined transfer 0.031222.
- Runtime 704.474929 ms, runtime inflation -0.020473, mean novelty 0.0, and complexity 0.0.
- Pareto efficient: False.

phase8_llm_static_selector

- Execution mode static_selector with replay none.
- Final held-out gap 0.217506, selector regret 0.145246, runtime-adjusted gap 0.217506.
- Family transfer 0.024668 and combined transfer 0.095713.
- Runtime 170.428671 ms, runtime inflation -0.76303, mean novelty 0.0, and complexity 0.62.
- Pareto efficient: True.
- Controller signature: annealed_multi_start:cluster_first_local_search:farthest_insertion_two_opt:nearest_neighbor_multistart:stag3:cand2:restart1.
- Controller rule summary: {"acceptance_bias": 0.006, "candidate_limit_offset": 2, "clustered_heuristic": "cluster_first_local_search", "corridor_heuristic": "limited_three_opt", "default_heuristic": "farthest_insertion_two_opt", "description": "Deterministic instance-adaptive selector over frozen TSP heuristic portfolio using structure descriptors (cluster, grid, bottleneck, corridor, NN-trap) and fixed bounded tuning/scheduling knobs.", "failed_perturbation_threshold": 2, "grid_like_heuristic": "candidate_pruned_two_opt", "low_improvement_threshold": 0.05, "name": "det_tsp_portfolio_oracleish_v1", "nearest_neighbor_trap_heuristic": "edge_preserving_restart", "random_like_heuristic": "annealed_multi_start", "restart_offset": 1, "stagnation_switch_heuristic": "nearest_neighbor_multistart", "stagnation_threshold": 3, "temperature_scale": 1.15, "time_budget_trigger": 0.65, "two_cluster_bottleneck_heuristic": "farthest_insertion_two_opt"}

phase8_llm_evolved_adaptive_controller

- Execution mode adaptive_controller with replay none.
- Final held-out gap 0.086887, selector regret 0.014627, runtime-adjusted gap 0.15705.
- Family transfer 0.106009 and combined transfer 0.098964.
- Runtime 1299.970086 ms, runtime inflation 0.807524, mean novelty 0.0, and complexity 0.7175.
- Pareto efficient: False.
- Controller signature: annealed_multi_start:cluster_first_local_search:edge_preserving_restart:annealed_multi_start:stag3:cand-2:restart1.
- Controller rule summary: {"acceptance_bias": 0.006, "candidate_limit_offset": -2, "clustered_heuristic": "cluster_first_local_search", "corridor_heuristic": "limited_three_opt", "default_heuristic": "farthest_insertion_two_opt", "description": "Deterministic instance-adaptive controller selecting from a frozen TSP heuristic portfolio. Uses interpretable structure cues (clusteredness, grid-likeness, bottleneck/two-cluster, corridor/elongation, and nearest-neighbor trap) plus online stagnation signals ", "failed_perturbation_threshold": 2, "grid_like_heuristic": "candidate_pruned_two_opt", "low_improvement_threshold": 0.09, "name":

"deterministic_instance_adaptive_tsp_hh_v1", "nearest_neighbor_trap_heuristic": "farthest_insertion_two_opt", "random_like_heuristic": "annealed_multi_start", "restart_offset": 1, "stagnation_switch_heuristic": "annealed_multi_start", "stagnation_threshold": 3, "temperature_scale": 1.15, "time_budget_trigger": 0.5, "two_cluster_bottleneck_heuristic": "edge_preserving_restart"}

phase8_llm_evolved_controller_diversity_failure_replay

- Execution mode adaptive_controller with replay diversity_failure.
- Final held-out gap 0.084733, selector regret 0.012473, runtime-adjusted gap 0.239104.
- Family transfer 0.090132 and combined transfer 0.088143.
- Runtime 2029.476943 ms, runtime inflation 1.821856, mean novelty 0.647902, and complexity 0.62.
- Pareto efficient: False.
- Controller signature: nearest_neighbor_multistart:cluster_first_local_search:edge_preserving_restart:annealed_multi_start:stag4:cand-1:restart2.
- Controller rule summary: {"acceptance_bias": 0.007, "candidate_limit_offset": -1, "clustered_heuristic": "cluster_first_local_search", "corridor_heuristic": "limited_three_opt", "default_heuristic": "farthest_insertion_two_opt", "description": "Deterministic instance-adaptive hyper-heuristic over a frozen portfolio; structure-driven core selection with conservative stagnation switching to reduce thrashing under hard/difficult replay modes.", "failed_perturbation_threshold": 3, "grid_like_heuristic": "candidate_pruned_two_opt", "low_improvement_threshold": 0.12, "name": "interpretable_instance_adaptive_tsp_hh_v2", "nearest_neighbor_trap_heuristic": "farthest_insertion_two_opt", "random_like_heuristic": "nearest_neighbor_multistart", "restart_offset": 2, "stagnation_switch_heuristic": "annealed_multi_start", "stagnation_threshold": 4, "temperature_scale": 1.3, "time_budget_trigger": 0.65, "two_cluster_bottleneck_heuristic": "edge_preserving_restart"}

phase8_full_solver_evolution

- Execution mode full_solver with replay none.
- Final held-out gap 0.099035, selector regret 0.026775, runtime-adjusted gap 0.099035.
- Family transfer 0.000881 and combined transfer 0.037043.
- Runtime 192.633443 ms, runtime inflation -0.732156, mean novelty 0.75, and complexity 0.76.
- Pareto efficient: True.

Judge Appendix

Conservative Interpretation of TSP Adaptive Heuristic Portfolio Results

1. **Primary Endpoint: Held-Out TSPLIB Gap**

- **Best fixed heuristic ("farthest_insertion_two_opt")** achieves a TSPLIB gap of **0.07226**, serving as the baseline for practical deployment.
- **Oracle selector** matches this gap (0.07226) but is a theoretical upper bound, not deployable.
- **Supervised ML selector** also achieves this same gap (0.07226) with slightly lower runtime than oracle.
- The **adaptive controllers** improve gap moderately but at the expense of increased runtime:
- "phase8_llm_evolved_adaptive_controller" gap = 0.0869 (worse than fixed heuristic) with much higher runtime (~1300 ms vs ~720 ms).

- "phase8_llm_evolved_controller_diversity_failure_replay" gap = 0.0847 but with very high runtime inflation and runtime (~2029 ms).
- The **full solver evolution** achieves a somewhat worse TSPLIB gap (0.099) but with significantly lower runtime (~193 ms), demonstrating a runtime-quality tradeoff.

2. **Selector Regret vs Oracle Portfolio**

- The **best fixed heuristic** and **supervised selector** have zero regret, indicating no underperformance relative to oracle on TSPLIB instances.
- Adaptive controllers have small selection regrets (around 0.01–0.03), but these do not translate to improved TSPLIB gaps.
- Small regret but increased runtime suggests they do not consistently improve solution quality vs cost.

3. **Runtime-Adjusted Gap**

- Despite the best TSPLIB gap, adaptive controllers have **worse runtime-adjusted gaps** compared to fixed or oracle selectors.
- The **static LLM-based selector** and **full solver evolution** are Pareto efficient, balancing quality and runtime better:
- Static selector has a higher gap (0.217) but *very low runtime* (170 ms), good for fast approximate solutions.
- Full solver evolution has lower gap (0.099) with runtime ~193 ms.
- Oracle and fixed heuristic have moderate runtime (~700 ms) but best or near-best gaps, representing a balanced point.

4. **Pareto Efficiency**

- **Pareto-efficient settings** are:
- **Oracle selector** (best gap, moderate runtime),
- **LLM static selector** (fast runtime but higher gap),
- **Full solver evolution** (moderate gap and low runtime).
- Adaptive controllers are **not Pareto-efficient** due to runtime inflation without gap improvements beyond fixed baseline.

5. **Cross-Family Transfer**

- Transfer gaps (performance on cross-family instances) are lowest for oracle and fixed heuristic (~0.026–0.031), showing stable transfer.
- Adaptive controllers show higher transfer gaps (~0.09–0.10), indicating potential overfitting or limited cross-family generalization.
- Static selector has intermediate transfer gap (~0.096), consistent with runtime and gap tradeoffs.

Summary of Findings

- **The best fixed heuristic ("farthest_insertion_two_opt") remains a strong baseline**, achieving the lowest held-out TSPLIB gap with moderate runtime.
- The **oracle selector confirms the theoretical upper bound** and is not deployable.

- **Supervised ML selector matches the fixed heuristic performance without runtime increase**, showing promise for interpretable instance-adaptive control.
- **Adaptive controllers, while interpretable and using instance structure plus stagnation signals, do not improve TSPLIB gap relative to fixed heuristic and incur high runtime inflation and transfer gap penalties**, limiting their practical utility at this stage.
- **Pareto analysis favors either the fixed/oracle-like selectors or low-runtime static selectors**, depending on runtime vs quality tradeoff preferences.
- The **full solver evolution approach offers a competitive balance between runtime and solution quality**, suggesting value in solver-level evolution over adaptive heuristic switching alone.

Recommendations

- Prioritize **deploying the fixed heuristic or supervised ML selector** for instance-adaptive hyper-heuristic control, as these approach oracle quality without runtime inflation.
- Investigate reducing runtime overhead and improving cross-family transfer in adaptive controllers before claiming benefits of interpretable instance-adaptive control.
- Avoid claims of novel algorithm discovery; current focus should remain on hyper-heuristic control over known heuristics with interpretable instance features.
- Consider further experiments with the full solver evolution approach given its Pareto efficiency and relatively low runtime.

This conservative interpretation respects the primary focus on held-out TSPLIB gap, runtime efficiency, selector regret, and transfer generalization without overclaiming advances beyond hyper-heuristic portfolio control.